

Question_2:

Code:

```
1  /*
2  Question_2: Sort horse names alphabetically, keeping royal gem-named horses first.
3  */
4
5  #include <iostream>
6  #include <vector>
7  #include <algorithm>
8  using namespace std;
9
10 string gems[] = {
11     "NONE", "Garnet", "Amethyst", "Aquamarine", "Diamond", "Emerald",
12     "Pearl", "Ruby", "Peridot", "Sapphire", "Tourmaline", "Topaz", "Lapis"
13 };
14 int rankGem(string s) {
15     int r = 0;
16     for (int i = 1; i < 13; i++)
17         if (s.find(gems[i]) != string::npos) r = max(r, i);
18     return r;
19 }
20 int main() {
21     vector<string> a;
22     string s;
23
24     while (getline(cin, s) && s != "END")
25         a.push_back(s);
26     sort(a.begin(), a.end(), [](string x, string y) {
27         int rx = rankGem(x), ry = rankGem(y);
28
29         if ((rx > 0) != (ry > 0))
30             return rx > 0;
31         if (rx != ry)
32             return rx > ry;
33         return x < y;
34     });
35     for (string s : a)
36         cout << s << '\n';
37     return 0;
38 }
```

Output:

Test Case_1:

```
Buttershy
Orangejack
Rubyanne
Rainbow Dash
Rarity
Pinkie Pie
Emerald Sunshine
Spike the Dragon
Misty Sapphire
END
Misty Sapphire
Rubyanne
Emerald Sunshine
Buttershy
Orangejack
Pinkie Pie
Rainbow Dash
Rarity
Spike the Dragon
PS C:\Users\Prabin Yadav\
```

Test Case_2:

```
Granny Smith
Discord
Spitfire
Granny Smith
Photo Finish
Pearl Peridot
Granny Smith
Princess Pearl
Soarine
END
Pearl Peridot
Princess Pearl
Discord
Granny Smith
Granny Smith
Granny Smith
Photo Finish
Soarine
Spitfire
PS C:\Users\Prabin Yadav\
```